The FREUID Challenge 2026

Technical Report — Nadhir Hasan

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Abstract

We present our solution to the FREUID Challenge 2026 identity-document fraud detection task. Our final model is a DINOv2-L Vision Transformer fine-tuned with low-rank adapters (LoRA, rank 16) and a per-patch multiple-instance-learning classification head, trained at 448×728 resolution. We train jointly on the official FREUID training set (5 document types) and 8 countries from the externally licensed IDNet dataset, using an annotation-driven synthetic fraud generator (portrait swap, cross-card region swap, text-field editing, and erase-and-retype) built on manually labeled per-type face and text-field bounding boxes. Critically, we select the training checkpoint using the exact competition metric (FREUID Score, combining AuDET and APCER@1%BPCER) evaluated on two IDNet countries held out entirely from training, rather than an in-domain FREUID validation split, which we found during development to be an unreliable predictor of leaderboard rank due to asset/persona duplication within the FREUID training distribution. Our submitted model scores [0.xxxxx] on the public leaderboard while maintaining substantially better cross-document-type generalization than a FREUID-only-trained variant of the same architecture. Code, weights, and a network-isolated Docker reproduction container are provided in the linked repository.

Competition: The FREUID Challenge 2026 (IJCAI-ECAI 2026)
Kaggle team: Nadhir Hasan
Code: https://github.com/nadhirhasan/FREUID-Challenge-2026 (commit b9473882f66a8baefce83980cf

1 Introduction

Identity-document fraud detection must generalize across a threat model that spans physical tampering, GenAI-driven digital manipulation, and print-and-capture ("analog hole") attacks, and across document types and issuing countries that are not exhaustively represented in any single training set. The FREUID Challenge formalizes this with a training set covering 5 document types and a scoring metric (FREUID Score, Section 5) that combines a global ranking measure (AuDET) with a strict low-false-alarm operating point (APCER@1%BPCER), penalizing solutions that overfit to easy global separability while failing at the operating point that matters for production KYC systems [3].

Our approach is built around a specific empirical finding from our own development process: a classifier trained *only* on the FREUID distribution can reach extremely low FREUID Score on the public leaderboard while being statistically indistinguishable from random (AUC ≈ 0.51) on document types it has never seen. We treat this as a direct proxy for the risk the private test set is designed to expose (it is drawn from 7 document types, including types not present in the public training data). We therefore deliberately trade some public-leaderboard score for measured cross-type generalization, by mixing in an external, permissively licensed ID-document dataset (IDNet) during training and selecting checkpoints on a held-out slice of that external data rather than on any FREUID-derived validation signal.

2 Method

2.1 Overview

Each input image is loaded at full resolution, converted to RGB, letterboxed (aspect-ratio-preserving resize + zero-padding) to a fixed 448×728 canvas, and normalized with standard ImageNet statistics. The backbone produces a token sequence; a per-patch head scores each spatial token independently and aggregates via a combination of top-k mean and attention-weighted pooling into a single image-level fraud logit. A sigmoid converts this logit to the submitted score in $[0, 1]$; no further post-processing or calibration is applied (Section 2.4).

2.2 Model architecture

Backbone: `vit_large_patch14_reg4_dinov2` (DINOv2-L [2] with register tokens), loaded via `timm`, Apache-2.0 licensed pre-trained weights, 24 transformer blocks, embedding dimension 1024. We attach a from-scratch, minimal LoRA implementation (rank 16, $\alpha = 32$, dropout 0.05; no third-party `peft` dependency) to the `qkv`, `proj`, `fc1`, and `fc2` linear layers of every transformer block; the backbone itself is frozen. On top of the token sequence we place a lightweight per-patch multiple-instance-learning (MIL) head: a 2-layer MLP (hidden size 512, GELU, dropout 0.3) scores every non-prefix token, and the image-level logit is the mean of (i) the top-10% most suspicious patch logits and (ii) an attention-weighted sum over all patch logits, each contributing equally. This design was chosen to push the model toward *local*, document-type-agnostic forgery cues (blend seams, splice boundaries, compression discontinuities) rather than global, country-specific layout statistics, which we found improves transfer to unseen document types. Trainable parameters: LoRA adapters + head only (≈7M of ≈304M total). Training/inference batch size: 6 (accumulated to an effective batch of 24) / 16–32 (inference only, no gradient state).

2.3 Training objective and procedure

Loss: Focal BCE ($\alpha = 0.25$, $\gamma = 2.0$) on the single whole-image logit. Optimizer: AdamW, two parameter groups (head lr 1×10^{-3}, LoRA lr 2×10^{-4}, weight decay 0.05/0 respectively), cosine learning-rate schedule with 5% linear warmup. Mixed precision (fp16 autocast + gradient scaler). Trained for 3 epochs over ≈111,000 images/epoch on a single NVIDIA RTX A4500 (20GB); ≈12–14 hours per fold at this configuration. We fine-tune the DINOv2 pre-trained weights via LoRA (frozen backbone) rather than training any component from scratch, except the LoRA adapters and the MIL head, which are randomly initialized.

Synthetic fraud generation. 25% of genuine training images (from both FREUID and IDNet sources) are converted to synthetic fraud examples each epoch, using one of four attack types, dispatched via manually annotated per-document-type face and text-field bounding boxes (Section 3.1): cross-card region/portrait swap (donor of the same document type), text-field edit (same-line content shifted and re-blended), erase-and-retype (ink removed via inpainting, new text rendered with a system font), and a generic self-blend fallback. Blending uses a mix of feathered alpha compositing and seamless (Poisson) cloning to avoid a single, learnable seam signature. This attack suite, compared against a simpler untargeted self-blend-only baseline under an otherwise identical recipe, improved our public leaderboard score by ≈43% in isolated ablation (Section 5).

2.4 Score calibration

No score calibration or thresholding is applied: the submitted label is $\sigma(\text{logit}) \in (0, 1)$ directly from the model's forward pass, identical at validation and test time. We rely on the FREUID Score's monotone invariance to score rescaling (it depends only on ranking and the genuine-score quantile that sets the 1%-BPCER threshold), so no train/test-specific calibration step is necessary or applied.

3 Data

3.1 FREUID training set

We use the official `train/` images and `train_labels.csv` metadata, stratified into 5 folds by (`type` $\times$ `label`) with a fixed seed; the submitted model uses fold 0 (all 5 FREUID document types are present in training — we do not hold out an entire document type, since the public/private test sets are drawn from the same underlying document-type distribution as the training set, per the challenge description). We additionally hand-annotated the portrait and text-field bounding boxes for every document type (5 FREUID + 10 IDNet countries) to drive the synthetic attack generator precisely rather than at random image locations.

3.2 External data and pre-trained models

- **DINOv2** [2] pre-trained ViT-L/14 weights, via `timm` (Apache-2.0 license).
- **IDNet** [1] (real, non-FREUID identity documents across 10 European countries), CC BY 4.0 license, publicly available at `https://arxiv.org/abs/2409.10472`. 8 countries (Spain, Albania, Azerbaijan, Finland, Greece, Latvia, Russia, Serbia) are mixed into training, capped at 56,000 images (balanced genuine/fraud) to roughly match the per-fold FREUID training volume. 2 further countries (Estonia, Slovakia) are held out entirely — never seen in training in any form — and used exclusively for checkpoint selection (Section 3.3).

Both artifacts are publicly available under licenses compatible with this competition's external-data policy ("any publicly available pre-trained model or dataset may be used, provided the license is compatible and the artifact is fully cited"); no proprietary or non-freely-accessible data was used. We deliberately avoided any model, checkpoint, or code released under a non-OSI-approved or non-commercial license (e.g. TruFor, ConvNeXt V2, DINOv3) anywhere in the training or inference pipeline for this submission.

3.3 Validation strategy

During development we found that in-domain FREUID validation (a stratified, disjoint 20% fold) is an *unreliable* predictor of public-leaderboard rank: in one controlled comparison, the fold with the numerically best in-domain validation score (by six orders of magnitude) scored *worse* on the public leaderboard than a fold with a substantially worse in-domain score. We traced this to heavy visual asset/persona reuse within the FREUID training distribution (near-duplicate faces and field values recur across nominally disjoint train/validation splits), which lets a model "pass" in-domain validation by recognizing recycled assets rather than genuine forgery cues. We therefore select the checkpoint to submit using the exact competition FREUID Score (Section 5) computed on the IDNet held-out slice described above (Section 3.1), which shares no assets with any training data by construction. Private test labels were never used, seen, or referenced during model selection.

4 Inference

At inference, each image is loaded, letterboxed to 448×728 (identical to training preprocessing, no cropping/tiling), and passed through the model in batches (no test-time augmentation or ensembling in the submitted configuration). The Docker entrypoint (`docker/prepare_submission.py`) discovers every image directly under `/data/`, runs inference, and writes `/submissions/submission.csv` with columns `id,label`. The container requires no network access at runtime (`-network none`); all model weights are baked into the image at build time. Inference auto-detects CUDA and falls back to CPU execution if unavailable.

5 Results

Table 1: Summary of main results (FREUID Score; lower is better).

Split / Leaderboard	FREUID ↓	Notes
IDNet held-out (selection metric)	0.1028	Selected checkpoint (epoch [X])
Public leaderboard	[0.xxxxx]	Selected final submission
Private (if known)	–	Not available at report time

Ablation — attack realism. Holding the architecture, data split, and all other hyperparameters fixed, replacing an untargeted self-blend-only synthetic-fraud generator with the annotation-driven attack suite (Section 2) improved the public leaderboard score from 0.00169 to 0.00096 (a FREUID-only-trained variant of the model, not the submitted generalist configuration).
Ablation — data mixing for generalization. A FREUID-only variant of the same architecture/recipe reached FREUID Score 0.00096 on the public leaderboard but 0.9662 (AUC ≈0.51, i.e. chance level) on the IDNet held-out set. Mixing in IDNet training data (this submission) reaches 0.1028 on the same held-out set — an order-of-magnitude improvement in measured cross-document-type generalization — at a cost of public-leaderboard score (0.00676), which we consider the correct trade given the private test set's document-type composition.

6 Reproducibility

- **Repository:** https://github.com/nadhirhasan/FREUID-Challenge-2026, commit `b9473882f66a8bae`
- **Docker:** `docker build -t freuid-repro:local .` from the repository root; weights are baked in at `/models/cv4_fold0.pt` (Git LFS in the repository).
- **Dependencies:** Python 3.11, PyTorch 2.6.0 (+cu124), timm 1.0.27, opencv-python-headless 4.10.0.84 — see `docker/requirements.txt` for the exact pinned inference environment (training used the same versions).
- **Runtime:** single RTX A4500 (20GB), ≈20 images/s at inference (batched, GPU); falls back to CPU automatically if no GPU is exposed to the container.
- **Randomness:** training uses unseeded per-worker RNGs for augmentation only (does not affect the frozen, deterministic inference path); inference is fully deterministic given fixed weights.
- **Network:** inference requires `-network none`; no downloads occur at container runtime. All dependencies and weights are installed/copied at build time.

7 Team and contributions

[FILL IN: team members and roles]

Acknowledgments

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References

[1] Hong Guo, Xuan Sun, Yu Liu, Xun Yang, and Xin Wang. IDNet: A novel dataset for identity document analysis and fraud detection research. https://arxiv.org/abs/2409.10472, 2024. CC BY 4.0 license; 8 training countries (ESP, ALB, AZE, FIN, GRC, LVA, RUS, SRB) and 2 held-out countries (EST, SVK) used for checkpoint selection in this work.

[2] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. DINOv2: Learning robust visual features without supervision. https://arxiv.org/abs/2304.07193, 2023. Apache-2.0 license; ViT-L/14 backbone used via timm.

[3] Ivan Relić, Vincenzo D'Elia, Stefano Bortoli, Radu Tudoran, Hristina Nedyalkova, Mihaela Bošnjak, Stanislav Pavlić, Tin Mavračić, Darin Dašić, Marin Kačan, Jerko Šegvić, Paolo Čerić, Filip Šoprun, Matej Miočić, Marin Oršić, Lorenzo Vaiani, and Paolo Garza. The FREUID challenge 2026: 1st competition on identity documents fraud detection. https://freuid2026.microblink.com/, 2026. IJCAI-ECAI 2026 competition.

A Hyperparameters

Hyperparameter	Value
Image size	448×728 (letterboxed)
Batch size (train)	6 (accum 4, effective 24)
Batch size (eval)	8
Learning rate (head)	1×10^{-3}
Learning rate (LoRA)	2×10^{-4}
Weight decay	0.05 (head) / 0.0 (LoRA)
LoRA rank / α / dropout	16 / 32 / 0.05
Epochs	3
Synthetic-attack rate (sbi)	0.25
IDNet training cap	56,000 (balanced)
IDNet held-out (selection) size	4,000 (balanced, 2 countries)
Seed (fold split)	42

B Additional ablations

See Section 5 for the two controlled single-lever ablations (attack realism; FREUID-only vs. FREUID+IDNet data mixing) that motivated the final configuration.